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Number Theory

An Original Olympiad Practice Set

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Original content. These 8 problems were written fresh for this set — they are **not** copied from Titu Andreescu's books or any competition. Eight original olympiad-style number theory problems spanning divisibility, modular arithmetic, multiplicative orders and primitive roots, p-adic valuation (Lifting the Exponent), and classic structure results. Each rewards a clean insight — periodicity towers, residue pairing, the order-divides-exponent lemma, LTE, and the smallest-prime-factor argument — rather than mechanical computation. Ordered easiest to hardest. Every closed-form answer was checked symbolically/numerically. Free to share for study.

Problems

1. DIFFICULTY 2/5 · MODULAR ARITHMETIC, EULER'S THEOREM, MULTIPLICATIVE ORDER, TOWER OF EXPONENTS

Find the last two digits of $7^{\left(7^7\right)}$.

2. DIFFICULTY 2/5 · P-ADIC VALUATION, LEGENDRE'S FORMULA, BOTTLENECK PRINCIPLE

Determine the largest integer k such that 12^k divides $100!$.

3. DIFFICULTY 4/5 · MULTIPLICATIVE ORDER, LIFTING THE EXPONENT, SMALLEST PRIME FACTOR ARGUMENT, STRONG INDUCTION

Find all positive integers n such that $n^2 \mid 2^n + 1$.

4. DIFFICULTY 3/5 · PRIMITIVE ROOTS, CYCLIC GROUP STRUCTURE, PAIRING OF INVERSES, EULER'S TOTIENT

Let p be an odd prime. Prove that the product of all primitive roots modulo p (taken among the residues $1, 2, \ldots, p-1$) is congruent to $1 \pmod p$ when $p > 3$, and to $-1 \pmod 3$ when $p = 3$.

5. DIFFICULTY 3/5 · POWER SUMS MODULO P, PRIMITIVE ROOT / ORDER CRITERION, FERMAT'S LITTLE THEOREM, GEOMETRIC SERIES

The number 2027 is prime. Find the remainder when

$$1^{2025} + 2^{2025} + 3^{2025} + \cdots + 2026^{2025}$$

is divided by 2027.

6. DIFFICULTY 4/5 · LIFTING THE EXPONENT (LTE), P-ADIC VALUATION, FACTORING A^N+B^N

Determine the largest integer N such that 3^N divides $2^{\left(3^7\right)} + 1$.

7. DIFFICULTY 4/5 · MULTIPLICATIVE ORDER, FERMAT NUMBERS, ORDER FORCES CONGRUENCE CLASS, FACTOR SEARCH

Consider $F = 2^{32} + 1 = 4294967297$. Prove that every prime divisor p of F satisfies $p \equiv 1 \pmod{64}$, and use this to find the smallest prime divisor of F.

8. DIFFICULTY 5/5 · MULTIPLICATIVE ORDER, EULER'S THEOREM, ORDER DIVIDES TOTIENT, STRUCTURAL / NO-COMPUTATION PROOF

Let $a \geq 2$ be an integer. Prove that for every integer $n \geq 1$,

$$n \mid \varphi(a^n - 1)\,,$$

where φ is Euler's totient function.

Solutions

1. Answer: 43

The last two digits of a number are its residue modulo 100. We compute $7^{(7^7)}$ mod 100. \par\textbf{Step 1: order of 7 modulo 100.} Since $\gcd(7,100)=1$ and $7^2=49$, $7^4=49^2=2401\equiv 1 \pmod{100}$. Thus the multiplicative order of 7 modulo 100 is 4 (it is not 1 or 2 since $7^2\equiv 49$). Therefore 7^E mod 100 depends only on E mod 4. \par\textbf{Step 2: reduce the exponent modulo 4.} We need $E=7^7$ mod 4. Since $7\equiv -1 \pmod 4$, we get $7^7\equiv(-1)^7=-1\equiv 3 \pmod 4$. \par\textbf{Step 3: assemble.} Hence $7^{(7^7)}\equiv 7^3 \pmod{100}$. Now $7^3=343\equiv 43 \pmod{100}$. \par The last two digits are $\boxed{43}$.

Second method. Use the Carmichael/Euler structure directly. Modulo 4: $7\equiv 3$, and 7^k mod 4 cycles with period 2 as $3,1,3,1,\ldots$, so since 7 is odd, $7^{(7^7)}\equiv 3\equiv -1 \pmod 4$. Modulo 25: $\lambda(25)=20$ and $7^4=2401=96\cdot 25+1\equiv 1 \pmod{25}$, so again only E mod 4 matters, giving $7^3=343\equiv 343-325=18 \pmod{25}$. Solve $x\equiv -1 \pmod 4,\ x\equiv 18 \pmod{25}$ by CRT: $x=43$ works ($43\equiv 3 \pmod 4$, $43\equiv 18 \pmod{25}$), giving 43.

2. Answer: 48

Write $12=2^2\cdot 3$. Then $12^k=2^{2k}\cdot 3^k$ divides 100! iff $2k\le v_2(100!)$ and $k\le v_3(100!)$, where v_p denotes the exponent of the prime p. \par\textbf{Legendre's formula} gives $v_p(n!)=\sum_{i\ge 1}\lfloor n/p^i\rfloor$. \par For $p=2$: $v_2(100!)=50+25+12+6+3+1=97$. \par For $p=3$: $v_3(100!)=33+11+3+1=48$. \par The two constraints are $2k\le 97$ (i.e. $k\le 48$, since k is an integer and $\lfloor 97/2\rfloor=48$) and $k\le 48$. The binding (smaller) bound is $k\le 48$ from the prime 3. \par Therefore the largest such k is $\boxed{48}$. The insight is that although 2 is far more abundant than 3, the 2^2 per copy of 12 means the supply of 2's suffices for 48 copies ($2\cdot 48=96\le 97$); the genuine bottleneck is the prime 3.

Second method. Compare supplies as ratios. Each 12 consumes two 2's and one 3. The number of available 2-pairs is $\lfloor 97/2\rfloor=48$; the number of available 3's is 48. The achievable count is the minimum, $\min(48,48)=48$.

3. Answer: $n \in \{1,\ 3\}$

Clearly $n = 1$ works ($1 \mid 3$) and $n = 3$ works ($9 \mid 9$). We show these are the only solutions. \par Suppose $n^2 \mid 2^n + 1$ with $n > 1$. Since $2^n + 1$ is odd, n is odd. Let p be the smallest prime divisor of n; then p is odd. \par\textbf{Step 1: $p = 3$.} From $p \mid 2^n + 1$ we get $2^n \equiv -1 \pmod p$, hence $2^{2n} \equiv 1 \pmod p$. Let $d = \mathrm{ord}_p(2)$. Then $d \mid 2n$ but $d \nmid n$ (else $2^n \equiv 1$, contradicting $2^n \equiv -1$ for odd p). Also $d \mid p - 1$ by Fermat. Now $d \mid 2n$ and $d \nmid n$ force $v_2(d) = v_2(2n) = 1$, so $d = 2m$ with $m \mid n$ odd. Since $m \mid n$ and $m \mid d \mid p - 1 < p$, every prime factor of m is a prime factor of n that is $< p$; by minimality of p, m has no prime factors, i.e. $m = 1$ and $d = 2$. Then $2^2 \equiv 1 \pmod p$, so $p \mid 3$, giving $p = 3$. Hence $3 \mid n$. \par\textbf{Step 2: bound the power of 3 via LTE.} Write $a = v_3(n) \geq 1$. Since n is odd, by the Lifting the Exponent lemma for $p = 3$,

$$v_3(2^n + 1) = v_3(2 + 1) + v_3(n) = 1 + a.$$

The hypothesis $n^2 \mid 2^n + 1$ requires $v_3(n^2) \leq v_3(2^n + 1)$, i.e. $2a \leq 1 + a$, so $a \leq 1$. Thus $a = 1$: exactly one factor of 3 divides n. \par\textbf{Step 3: no other prime divides n.} Suppose $n > 3$. Write $n = 3m$ with $m > 1$ and $\gcd(m, 3) = 1$ (by Step 2). Let q be the smallest prime factor of m; then $q > 3$, and q is the smallest prime factor of n other than 3. From $q \mid 2^n + 1$, repeat the order analysis: $e := \mathrm{ord}_q(2)$ satisfies $e \mid 2n$, $e \nmid n$, so $e = 2t$ with t odd, $t \mid n$. Each prime factor of t divides n and is $\leq t \mid e \mid q - 1 < q$, so it is a prime factor of n smaller than q — hence it must be 3. Thus $t \in \{1, 3\}$ (as $v_3(n) = 1$), giving $e \in \{2, 6\}$, so $q \mid 2^2 - 1 = 3$ or $q \mid 2^6 - 1 = 63 = 3^2 \cdot 7$. The only prime > 3 here is $q = 7$. But $e = 6$ and $7 \mid 2^n + 1$ would need $2^n \equiv -1 \pmod 7$; the powers of 2 mod 7 cycle $2, 4, 1$ and never equal $6 \equiv -1$. Contradiction. Hence no such q exists and $m = 1$. \par Therefore $n = 3$, and the complete solution set is $\boxed{\{1, 3\}}$.

Second method. After Step 1 establishes $3 \mid n$ and Step 2 (LTE) forces $v_3(n) = 1$, one can finish by a clean lifting argument: assume for contradiction $n > 3$ and take the second-smallest prime $q \mid n$. Since $2^{2n} \equiv 1 \pmod q$ but $2^n \equiv -1$, the order of 2 mod q is even, equal to $2t$ with $t \mid n$; minimality forces t to be built only from the already-found prime 3, so $\mathrm{ord}_q(2) \in \{2, 6\}$ and $q \mid 2^6 - 1$. Checking $q = 7$ directly fails since -1 is not a power of 2 mod 7. This contradiction gives $n = 3$.

4. Answer: Proof

For $p = 3$, the only primitive root is 2, and the product is $2 \equiv -1 \pmod 3$. Now assume $p > 3$. \par\textbf{Setup.} The group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$; fix a generator g. Every element is g^k for a unique $k \in \{0, 1, \ldots, p-2\}$, and g^k is a primitive root iff $\gcd(k, p-1) = 1$. There are $\varphi(p-1)$ primitive roots, indexed by the set $U = \{k : 1 \le k \le p-1,\ \gcd(k, p-1) = 1\}$. \par\textbf{Product of indices.} The product of all primitive roots is

$$P \equiv \prod_{k \in U} g^k = g^S, \qquad S = \sum_{k \in U} k.$$

\par\textbf{Symmetry of the exponent sum.} The set U is symmetric under $k \mapsto (p-1) - k$: indeed $\gcd(k, p-1) = \gcd((p-1) - k,\ p-1)$. Since $p > 3$, we have $p - 1 > 2$, so no element is fixed by this involution (a fixed point would need $k = (p-1)/2$, but then $\gcd(k, p-1) = (p-1)/2 > 1$, so it is not in U). Hence U splits into $\frac{1}{2}\varphi(p-1)$ pairs $\{k, (p-1) - k\}$, each summing to $p - 1$. Therefore

$$S = \frac{\varphi(p-1)}{2}\,(p-1).$$

\par\textbf{Conclusion.} Then $P \equiv g^S = \left(g^{p-1}\right)^{\varphi(p-1)/2} \equiv 1^{\varphi(p-1)/2} = 1 \pmod p$, since $g^{p-1} \equiv 1$ by Fermat. (Note $\varphi(p-1)$ is even for $p - 1 > 2$, so S is an integer multiple of $p - 1$.) This proves $P \equiv 1 \pmod p$ for $p > 3$. ■

Second method. Pair each primitive root with its inverse. If r is a primitive root then so is r^{-1}, and $r \cdot r^{-1} \equiv 1$. A root is its own inverse only if $r^2 \equiv 1$, i.e. $r \equiv \pm 1$; but 1 has order 1 and -1 has order 2, neither equal to $p - 1$ when $p > 3$. Hence the primitive roots pair off into self-inverse-free pairs whose product is 1, so the total product is $1 \pmod p$. For $p = 3$, $p - 1 = 2$ and the lone primitive root $2 \equiv -1$ is its own inverse, giving -1.

5. Answer: 0

Let $p = 2027$ (prime) and $k = 2025$; note $p - 1 = 2026$. We evaluate $S = \sum_{a=1}^{p-1} a^k \bmod p$. \par\textbf{Key lemma.} For a prime p and integer $k \geq 1$,

$$\sum_{a=1}^{p-1} a^k \equiv \begin{cases} -1 \pmod p, & (p-1) \mid k, \\ 0 \pmod p, & (p-1) \nmid k. \end{cases}$$

\par\textbf{Proof of lemma.} Let g be a primitive root mod p. As a ranges over $1, \ldots, p-1$, the residue a ranges over $g^0, g^1, \ldots, g^{p-2}$ in some order. Hence

$$S \equiv \sum_{j=0}^{p-2} (g^j)^k = \sum_{j=0}^{p-2} (g^k)^j.$$

This is a geometric series with ratio $t = g^k$. If $(p-1) \mid k$ then $t \equiv 1$ and the sum is $p - 1 \equiv -1$. If $(p-1) \nmid k$ then $t \not\equiv 1$ (since g has order $p-1$), and

$$S \equiv \frac{t^{p-1} - 1}{t - 1} \equiv \frac{1 - 1}{t - 1} = 0 \pmod p,$$

using $t^{p-1} = g^{k(p-1)} \equiv 1$ by Fermat. □ \par\textbf{Apply.} Here $p - 1 = 2026$ and $k = 2025$. Since $2026 \nmid 2025$ (indeed $0 < 2025 < 2026$), we are in the second case, so $S \equiv 0 \pmod{2027}$. \par The remainder is $\boxed{0}$.

Second method. Pair a with $p - a$. Since $k = 2025$ is odd, $(p-a)^k \equiv (-a)^k = -a^k \pmod p$. Thus $a^k + (p-a)^k \equiv 0$, and the $p - 1 = 2026$ terms split into 1013 such pairs (no term is self-paired since p is odd, so $a \neq p - a$). Each pair contributes 0, giving total $0 \pmod{2027}$. This pairing argument needs only that the exponent is odd.

6. Answer: 8

We must compute $v_3\left(2^{3^7} + 1\right)$, the exponent of 3 in the factorization. \par\textbf{LTE for sums.} The Lifting the Exponent lemma states: for an odd prime p with $p \mid a + b$ and $p \nmid a$, $p \nmid b$, and for an odd positive integer m,

$$v_p(a^m + b^m) = v_p(a + b) + v_p(m).$$

\par\textbf{Apply with $a = 2,\ b = 1,\ p = 3,\ m = 3^7$.} Here $3 \mid 2 + 1 = 3$, $3 \nmid 2$, $3 \nmid 1$, and $m = 3^7$ is odd. Hence

$$v_3\left(2^{3^7} + 1\right) = v_3(2 + 1) + v_3(3^7) = v_3(3) + 7 = 1 + 7 = 8.$$

\par Therefore 3^8 divides $2^{3^7} + 1$ but 3^9 does not, and the answer is $N = \boxed{8}$.

Second method. Prove the increment by induction without quoting LTE. Let $x_k = 2^{3^k} + 1$. Then $x_{k+1} = 2^{3^{k+1}} + 1 = \left(2^{3^k}\right)^3 + 1 = (x_k - 1)^3 + 1$. Expanding, $x_{k+1} = x_k\big((x_k - 1)^2 - (x_k - 1) + 1\big) = x_k\,(x_k^2 - 3x_k + 3)$. Since $3 \mid x_k$ (as $x_0 = 3$ and the recursion preserves divisibility by 3), the factor $x_k^2 - 3x_k + 3$ is $\equiv 3 \pmod 9$ (each of x_k^2 and $3x_k$ is divisible by 9 for $k \geq 1$, leaving 3), so v_3 of that factor is exactly 1. Thus $v_3(x_{k+1}) = v_3(x_k) + 1$. With $v_3(x_0) = v_3(3) = 1$, induction gives $v_3(x_k) = k + 1$; for $k = 7$ this is 8.

7. Answer: 641

\textbf{Step 1: a congruence for prime divisors.} Let p be a prime with $p \mid 2^{32}+1$. Then $2^{32} \equiv -1 \pmod{p}$, so squaring gives $2^{64} \equiv 1 \pmod{p}$. Let $d = \operatorname{ord}_p(2)$. Then $d \mid 64$, so d is a power of 2. But $d \nmid 32$: if $d \mid 32$ we would have $2^{32} \equiv 1$, contradicting $2^{32} \equiv -1 \pmod{p}$ (these differ since p is odd, $p \nmid 2$). The only divisor of 64 that does not divide 32 is 64 itself, so $d = 64$. \par By Fermat's little theorem $d \mid p-1$, hence $64 \mid p-1$, i.e. $p \equiv 1 \pmod{64}$. This proves the claim for every prime divisor. \par\textbf{Step 2: search the admissible candidates.} A prime factor must have the form $p = 64k+1$. Testing the primes of this form in increasing order: \par $65 = 5 \cdot 13$ (not prime); $129 = 3 \cdot 43$ (not prime); 193 prime — $193 \nmid F$; 257 prime — $257 \nmid F$; $321 = 3 \cdot 107,\ 385 = 5 \cdot 7 \cdot 11$ (not prime); 449 prime — $449 \nmid F$; $513 = 27 \cdot 19$ (not prime); 577 prime — $577 \nmid F$; 641 prime — and $641 \mid F$. \par Indeed $F = 4294967297 = 641 \cdot 6700417$, so the smallest prime divisor is $\boxed{641}$. \par (One can confirm $641 \mid 2^{32}+1$ elegantly: $641 = 5^4 + 2^4 = 5 \cdot 2^7 + 1$. From $5 \cdot 2^7 \equiv -1$ we get $5^4 \cdot 2^{28} \equiv 1$; and from $5^4 \equiv -2^4$ we get $-2^4 \cdot 2^{28} = -2^{32} \equiv 1$, i.e. $2^{32} \equiv -1 \pmod{641}$.)

Second method. The slick divisibility check by Euler/Fermat's identity: $641 = 2^4 + 5^4 = 1 + 5 \cdot 2^7$. The second form gives $5 \cdot 2^7 \equiv -1 \pmod{641}$, so $(5 \cdot 2^7)^4 \equiv 1$, i.e. $5^4 \cdot 2^{28} \equiv 1$. The first form gives $5^4 \equiv -2^4 \pmod{641}$. Substituting, $-2^4 \cdot 2^{28} = -2^{32} \equiv 1$, hence $2^{32} \equiv -1$ and $641 \mid 2^{32}+1$. Combined with Step 1's lower bound (any prime factor is at least 193, and one checks $193, 257, 449, 577$ fail), 641 is the least prime factor.

8. Answer: Proof

Set $m = a^n - 1$. For $n = 1$, $\varphi(a-1)$ is divisible by 1 trivially, so assume $n \geq 2$; then $m = a^n - 1 \geq a^2 - 1 \geq 3$. \par\textbf{Step 1: a is a unit modulo m.} Since $a^n \equiv 1 \pmod{m}$ by construction ($m \mid a^n - 1$), a is invertible modulo m; equivalently $\gcd(a, m) = 1$ (any common prime factor would divide a^n and $a^n - 1$, hence 1). So a lies in the multiplicative group $(\mathbb{Z}/m\mathbb{Z})^\times$, which has order $\varphi(m)$. \par\textbf{Step 2: the order of a modulo m is exactly n.} Let $d = \operatorname{ord}_m(a)$. From $a^n \equiv 1$ we get $d \mid n$. Conversely, suppose $d < n$. Then $a^d \equiv 1 \pmod{m}$ means $m \mid a^d - 1$, so

$$a^n - 1 = m \leq a^d - 1.$$

But $a \geq 2$ and $d < n$ give $a^d - 1 < a^n - 1$, a strict inequality, contradicting $a^n - 1 \leq a^d - 1$. (We used that $a^d - 1 \geq 1 > 0$, so divisibility forces $m \leq a^d - 1$.) Hence no $d < n$ works, and $d = n$. \par\textbf{Step 3: Lagrange / Euler.} The order of any element of a finite group divides the group's order (Lagrange); equivalently, $\operatorname{ord}_m(a) \mid \varphi(m)$ by Euler's theorem ($a^{\varphi(m)} \equiv 1$). Therefore

$$n = \operatorname{ord}_m(a) \ \big|\ \varphi(m) = \varphi(a^n - 1).$$

\par This holds for all $n \geq 1$ and all $a \geq 2$. ■ \par\textbf{Remark.} The crux is recognizing that $a^n - 1$ is built precisely so that a has order exactly n modulo it — minimality of the exponent comes for free from the size comparison $a^d - 1 < a^n - 1$. No factorization of $a^n - 1$ is needed.

Second method. Frame it via cyclotomic structure: $a^n - 1 = \prod_{d \mid n} \Phi_d(a)$ where Φ_d is the d-th cyclotomic polynomial. A prime p dividing $\Phi_n(a)$ (and not dividing n) satisfies $\operatorname{ord}_p(a) = n$, forcing $n \mid p-1$; the primitive prime divisor existence (Zsygmondy, for $n \geq 2$, $(a,n) \neq (2,6)$) then exhibits such a p with $n \mid p - 1 \mid \varphi(a^n - 1)$. However the order argument above is cleaner and fully self-contained, handling all cases (including $(a,n) = (2,6)$) uniformly, since it only needs that a itself has order n modulo $a^n - 1$, which is elementary.